

# Sai Prasanna Raman

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## Profile

PhD student researching to enable adaptive, curious, **generally capable robots**. Currently investigating causal world models and reinforcement learning for robotics.

## Work Experience

### PhD Student,

Aug 2025 – Present

University of Tübingen/Max Planck Institute for Intelligent Systems

- Research on **Causal World Models** advised by Prof. Georg Martius and Prof. Bernhard Schölkopf

### Research Assistant, Machine Learning Lab, University of Freiburg

Nov 2023 – Jul 2025

- Successful research in analyzing and improving **world models** to allow **Zero-Shot Generalization** in Contextual RL.
- As part of my ongoing thesis, I am creating simple **RL benchmarks** for evaluating the performance and **state estimation** capabilities of world models under **partial observability**.
- Working on methods to improve **the performance of model-based RL** in partially observable environments.

### Research Assistant, Robot Learning Lab, University of Freiburg

Jan 2022 – Oct 2023

- Developed **uncertainty-aware semantic mapping** to allow sample efficient reinforcement learning in embodied indoor navigation tasks.
- Created new environments and base-policies in **NVIDIA Isaac** simulator to facilitate hierarchical reinforcement learning in mobile manipulation tasks.
- Research and engineering on using world models for **cross-embodiment generalization**.

### Staff Machine Learning Scientist, Agara Labs

2021

- Significantly improved the accuracy of a **hierarchical text classification** system for voice support using large language models (LLMs), reducing operational time for customer support representatives.

### Machine Learning Engineer/Member Leadership Staff, Zoho Corporation

2017 – 2021

#### Grammatical Error Correction

- Developed a deep learning-based **multi-lingual grammatical** and **contextual language assistance** system for Zoho Writer, which has proofread and helped correct nearly a billion (and counting) user-written sentences
- Improved the accuracy of the system through the application of **semi-supervised learning**, **data augmentation**, architectures with **better inductive bias**, and **transfer learning**. Scaled up the model inference using **quantization**, **distillation**, and **pruning** techniques.
- Led the release of grammar correction for the Spanish using **LLM-based Zero-Shot approach** and initiated active research on adding new languages to assist diverse language writers with proofreading needs.
- **Mentored** junior engineers in converting research in fine-grained **grammatical error classification** to production-ready models.
- Contributed back to **open source** NLP projects such as AllenNLP and FairSeq.

#### NLP Suite

- Made critical architecture choices for multiple services in Zoho's NLP suite, such as **translation**, **question answering**, and **language detection**.
- Designed a prototype scalable semantic search system by incorporating advances in **approximate nearest neighbours** search and **semantic vectors** from deep neural nets.

### Software Engineer, ManageEngine, Zoho Corporation

2015 – 2017

- Spearheaded the development of a **remote desktop client** for the GNU/Linux platform, expanding the platforms that can be remotely managed by users of Desktop Central and Zoho Assist.
- Developed the **iOS app** for network management and **led the team** responsible for building the Android app for the same. These apps provided users with secure and easy access to network resources on-the-go, improving their productivity and efficiency.

## Publications

### One Does Not Simply Estimate State: Comparing Model-based and Model-free Reinforcement Learning on the Partially Observable MordorHike Benchmark,

Sep 2025

In *Eighteenth European Workshop on Reinforcement Learning*, 2025

Sai Prasanna, André Biedenkapp, and Raghu Rajan

- We introduce a simple but challenging Partially Observable Markov Decision Process **benchmark** and compare the performance of model-based and model-free RL algorithms.

### Dreaming of Many Worlds: Learning Contextual World Models Aids Zero-Shot Generalization,

Aug 2024

*Reinforcement Learning Journal*, 3, pp. 1317–1350, 2024

Sai Prasanna, Karim Farid\*, Raghu Rajan, André Biedenkapp

- We introduce the contextual recurrent state-space model (cRSSM) and enhance the ZSG of RL policies trained using world models.

Perception Matters: Enhancing Embodied AI with Uncertainty-Aware Semantic Segmentation, ISRR 2024

Sai Prasanna\*, Daniel Honerkamp\*, Kshitij Sirohi\*, Tim Welschehold, Wolfram Burgard, Abhinav Valada

Improved success rates of the embodied **multi-object search** in unseen indoor environments by incorporating **uncertainty aware perception** module.

When BERT Plays the Lottery, All Tickets Are Winning!, EMNLP 2020

Sai Prasanna, Anna Rogers, Anna Rumshisky

Analyzed pruned sub-networks in BERT in the context of fine-tuning. Used **pruning** as a **novel approach** for **interpretability**.

Zoho at SemEval-2019 Task 9: Semi-supervised domain adaptation using tri-training for suggestion mining, SemEval workshop @ NAACL 2019

Sai Prasanna and Ananda Seelan

Incorporated transfer learning and **semi-supervised learning** for **domain adaptation** reaching **third place** in the leaderboard for the domain transfer.

Dec 202420202019

Scholarships

ELIZA Master's Scholarship, Konrad Zuse School of Excellence in Learning and Intelligent Systems

Among the select (forty) M.Sc CS candidates across Germany awarded the funding to do research in AI for two years.

Oct 2023

Research Projects

Augmented Dreams: Data Augmentation & Self-supervised learning in Model-Based Reinforcement Learning, Deep Learning Lab - University of Freiburg

Studied the effects of **data augmentation** and **self-supervised learning** auxiliary losses for improving **sample-efficiency** of model-based RL algorithm (Dreamer).

2022

DEHB-WS: Joint Architecture and Hyperparameter Search with Weight Sharing, AutoML Lab - University of Freiburg

Introduced a new efficient method for **joint architecture** and **hyperparameter search**.

2022

Evaluating Zero-th and First-Order MPC Methods with a World Model, Neurorobotics Lab - University of Freiburg

Research on improving the **online planning** performance of the a world model by exploiting **gradient** information in conjunction with black box optimization method.

2022

Imitation Learning in MPC using Differentiable Convex Optimization, Systems Control and Optimization Lab - University of Freiburg

Applied **differentiable convex optimization** to improve **imitation learning** compared to system identification for Model Predictive Control (MPC).

2023

Skills

| Research Areas                                                   | Relevant Courses                                               | Programming Languages               | Libraries/Frameworks                                      |
|------------------------------------------------------------------|----------------------------------------------------------------|-------------------------------------|-----------------------------------------------------------|
| Reinforcement Learning, World Models, Embodied AI, Deep learning | Reinforcement Learning, Deep Learning, Optimal Control, AutoML | Python, Swift, Javascript, Go, Rust | Pytorch, Jax, NVIDIA Isaac Sim, Stablebaselines, AllenNLP |

Education

Master of Science - Computer Science, University of Freiburg

Specialization in Artificial Intelligence

GPA: 1.2

Oct 2021 – Apr 2025 | Freiburg, Germany

Scale: 1.0 (best) - 4.0 (passing)

Bachelor of Engineering - Computer Science, Anna University

CGPA: 8.24

Jun 2011 – Apr 2015 | Chennai

Scale: 10.0 (best) - 5.0 (passing)